RaceDay - Part 1 Design Rationale

Programming 2B | Module Code PROG6212 | Supporting planning document

1. Relational data model

The model uses six entities: Users, Events, Categories, EventCategories, Enrolments and Results. Users stores identity and role information. Events is owned by an Organiser. EventCategories resolves the Event-Category many-to-many relationship and stores event-specific entry rules. Enrolments links a Participant to an EventCategory, while Results records an optional result for an enrolment.

Primary and foreign keys identify rows and maintain referential integrity. UNIQUE and CHECK constraints protect data integrity by preventing duplicate values and restricting invalid domain values (Microsoft, 2026b; Microsoft, 2026c).

2. API planning

The endpoint plan uses resource-oriented routes and standard HTTP methods. GET is used for retrieval, POST for creation, PUT for updates and DELETE for removal in accordance with HTTP semantics (IETF, 2022). The resource-oriented approach is consistent with the REST architectural style (Fielding, 2000).

Role requirements are explicit in the endpoint table. This reflects the distinction between authentication and authorization and provides a clear basis for API-level role enforcement (Microsoft, 2026a).

3. Consistency controls

The same entity names and relationships are used in the ERD, endpoint plan and SQL script. The SQL script contains two Organisers, two Participants, three Events, event categories, sample enrolments and sample results. Event records include an optional route-information URL; live weather is treated as dynamic external information rather than a stored relational entity.

References

Fielding, R.T. (2000) *Architectural Styles and the Design of Network-based Software Architectures*. PhD dissertation. University of California, Irvine. Available at: https://ics.uci.edu/~fielding/pubs/dissertation/ (Accessed: 22 September 2026).

IETF (2022) *RFC 9110: HTTP Semantics*. Available at: https://www.rfc-editor.org/rfc/rfc9110.html (Accessed: 22 September 2026).

Microsoft (2026a) *Introduction to authorization in ASP.NET Core*. Microsoft Learn. Available at: https://learn.microsoft.com/en-us/aspnet/core/security/authorization/introduction (Accessed: 22 September 2026).

Microsoft (2026b) *Primary and foreign key constraints - SQL Server*. Microsoft Learn. Available at: https://learn.microsoft.com/en-us/sql/relational-databases/tables/primary-and-foreign-key-constraints (Accessed: 22 September 2026).

Microsoft (2026c) *Unique constraints and check constraints - SQL Server*. Microsoft Learn. Available at: https://learn.microsoft.com/en-us/sql/relational-databases/tables/unique-constraints-and-check-constraints (Accessed: 22 September 2026).